

Algebra Foundations

A variable is a symbol that represents an unknown or changing number. An expression combines numbers, variables, and operations, while an equation states that two expressions are equal.

To solve a linear equation, keep both sides balanced. Apply the same operation to each side. For example, $3x + 5 = 20$ becomes $3x = 15$ after subtracting 5, then $x = 5$ after dividing by 3.

The distributive property states $a(b + c) = ab + ac$. Like terms have the same variable parts and can be combined: $4x + 2x = 6x$. Unlike terms such as $4x$ and $2y$ cannot be combined.

A ratio compares quantities by division. A proportion states that two ratios are equal. Cross multiplication can solve a proportion when the denominators are nonzero.